

1 Problem 1: Packetization Delay, Transmission Delay, and Propagation Delay – VoIP

Problem Statement

Host A converts analog voice into a digital bit stream at a rate of 64 kbps. Host A groups the bits into 56-byte packets. There is one link between Hosts A and B with transmission rate 10 Mbps and propagation delay 10 ms.

As soon as Host A gathers one complete packet, it sends it to Host B. As soon as Host B receives the complete packet, it converts the packet back to an analog signal.

Determine the total time elapsed from the creation of a bit at Host A until that bit is decoded at Host B.

The total delay experienced by a bit consists of three components:

1. Packetization delay
2. Transmission delay
3. Propagation delay

We calculate each component separately.

Step 1: Packet Size

The packet contains 56 bytes.

Since

$$1 \text{ byte} = 8 \text{ bits},$$

the packet size in bits is

$$L = 56 \times 8.$$

Therefore,

$$\boxed{L = 448 \text{ bits}}.$$

Step 2: Packetization Delay

The voice source generates bits at

$$R_{\text{voice}} = 64 \text{ kbps} = 64,000 \text{ bits/sec.}$$

The packet contains 448 bits.

Therefore, the time required for Host A to generate all 448 bits needed to form one packet is

$$T_{\text{packetization}} = \frac{L}{R_{\text{voice}}}.$$

Substituting the values,

$$T_{\text{packetization}} = \frac{448}{64,000}.$$

Hence,

$$T_{\text{packetization}} = 0.007 \text{ sec.}$$

Therefore,

$$\boxed{T_{\text{packetization}} = 7 \text{ ms}}.$$

This means that a bit created at the very beginning of a packet has to wait 7 ms for the remaining bits of the packet to be generated.

Step 3: Transmission Delay

The transmission rate of the link is

$$R = 10 \text{ Mbps} = 10 \times 10^6 \text{ bits/sec.}$$

The packet contains 448 bits.

Therefore,

$$T_{\text{transmission}} = \frac{L}{R}.$$

Thus,

$$T_{\text{transmission}} = \frac{448}{10 \times 10^6}.$$

Hence,

$$T_{\text{transmission}} = 0.0000448 \text{ sec.}$$

Converting into milliseconds,

$$0.0000448 \times 1000 = 0.0448 \text{ ms.}$$

Therefore,

$$\boxed{T_{\text{transmission}} = 0.0448 \text{ ms}}.$$

Step 4: Propagation Delay

The propagation delay is directly given as

$$\boxed{T_{\text{propagation}} = 10 \text{ ms}}.$$

Step 5: Total Delay

For the first bit generated in a packet, the total delay is

$$T_{\text{total}} = T_{\text{packetization}} + T_{\text{transmission}} + T_{\text{propagation}}.$$

Substituting,

$$T_{\text{total}} = 7 + 0.0448 + 10$$

milliseconds.

Therefore,

$$T_{\text{total}} = 17.0448 \text{ ms.}$$

Hence,

$$T_{\text{total}} = 17.0448 \text{ ms}$$

or approximately

$$T_{\text{total}} \approx 17.045 \text{ ms.}$$

Final Answer

$$\text{Total delay} = 17.0448 \text{ ms} \approx 17.045 \text{ ms}$$

2 Problem 2: End-to-End Delay over Multiple Links

Problem Statement

A packet of length L travels from an end system A to a destination through three links and two packet switches.

For link i :

- d_i = length of the link
- s_i = propagation speed
- R_i = transmission rate

Each packet switch introduces a processing delay d_{proc} . There are no queuing delays. First, derive the total end-to-end delay in terms of d_i , s_i , R_i , and L . Then calculate the numerical value for the given parameters.

The total end-to-end delay consists of:

1. Transmission delay on each link
2. Propagation delay on each link
3. Processing delay at each packet switch

There are three links and two packet switches.

Step 1: Transmission Delay

For a packet of length L transmitted at rate R_i , the transmission delay on link i is

$$d_{\text{trans},i} = \frac{L}{R_i}.$$

Therefore, for all three links,

$$T_{\text{trans}} = \frac{L}{R_1} + \frac{L}{R_2} + \frac{L}{R_3}.$$

Thus,

$$T_{\text{trans}} = \sum_{i=1}^3 \frac{L}{R_i}.$$

Step 2: Propagation Delay

Propagation delay on link i is

$$d_{\text{prop},i} = \frac{d_i}{s_i}.$$

Therefore,

$$T_{\text{prop}} = \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}.$$

Hence,

$$T_{\text{prop}} = \sum_{i=1}^3 \frac{d_i}{s_i}.$$

Step 3: Processing Delay

There are two packet switches.

Each introduces a processing delay of

$$d_{\text{proc}}.$$

Therefore,

$$T_{\text{proc}} = 2d_{\text{proc}}.$$

Step 4: Total End-to-End Delay

The total delay is

$$T_{\text{end-to-end}} = T_{\text{trans}} + T_{\text{prop}} + T_{\text{proc}}.$$

Therefore,

$$T_{\text{end-to-end}} = \frac{L}{R_1} + \frac{L}{R_2} + \frac{L}{R_3} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + 2d_{\text{proc}}$$

This is the required general expression.

2.1 Numerical Calculation

Given:

$$L = 1500 \text{ bytes.}$$

Convert into bits:

$$L = 1500 \times 8 = 12,000 \text{ bits.}$$

The propagation speed on all links is

$$s_1 = s_2 = s_3 = 2.5 \times 10^8 \text{ m/s.}$$

The transmission rate on all links is

$$R_1 = R_2 = R_3 = 2.5 \text{ Mbps} = 2.5 \times 10^6 \text{ bits/sec.}$$

Processing delay at each switch is

$$d_{\text{proc}} = 3 \text{ ms.}$$

The link lengths are:

$$d_1 = 5000 \text{ km,}$$

$$d_2 = 4000 \text{ km,}$$

$$d_3 = 1000 \text{ km.}$$

Convert kilometers into meters:

$$d_1 = 5 \times 10^6 \text{ m,}$$

$$d_2 = 4 \times 10^6 \text{ m,}$$

$$d_3 = 1 \times 10^6 \text{ m.}$$

Step 1: Transmission Delay on Each Link

Since all three links have the same rate,

$$T_{\text{trans},i} = \frac{12,000}{2.5 \times 10^6}.$$

Therefore,

$$T_{\text{trans},i} = 0.0048 \text{ sec} = 4.8 \text{ ms.}$$

There are three links:

$$T_{\text{trans}} = 3(4.8) = 14.4 \text{ ms.}$$

Thus,

$$\boxed{T_{\text{trans}} = 14.4 \text{ ms}}.$$

Step 2: Propagation Delay

For the first link,

$$T_{\text{prop},1} = \frac{5 \times 10^6}{2.5 \times 10^8}.$$

Thus,

$$T_{\text{prop},1} = 0.02 \text{ sec} = 20 \text{ ms}.$$

For the second link,

$$T_{\text{prop},2} = \frac{4 \times 10^6}{2.5 \times 10^8} = 0.016 \text{ sec} = 16 \text{ ms}.$$

For the third link,

$$T_{\text{prop},3} = \frac{1 \times 10^6}{2.5 \times 10^8} = 0.004 \text{ sec} = 4 \text{ ms}.$$

Therefore,

$$T_{\text{prop}} = 20 + 16 + 4 = 40 \text{ ms}.$$

Hence,

$$\boxed{T_{\text{prop}} = 40 \text{ ms}}.$$

Step 3: Processing Delay

There are two packet switches.

Therefore,

$$T_{\text{proc}} = 2(3 \text{ ms}) = 6 \text{ ms}.$$

Hence,

$$\boxed{T_{\text{proc}} = 6 \text{ ms}}.$$

Step 4: Total Delay

Therefore,

$$T_{\text{end-to-end}} = T_{\text{trans}} + T_{\text{prop}} + T_{\text{proc}}.$$

Substituting,

$$T_{\text{end-to-end}} = 14.4 + 40 + 6.$$

Hence,

$$\boxed{T_{\text{end-to-end}} = 60.4 \text{ ms}}.$$

Equivalently,

$$\boxed{T_{\text{end-to-end}} = 0.0604 \text{ sec}}.$$

3 Problem 3: Store-and-Forward Delay versus Immediate Forwarding

Problem Statement

Consider Problem 2, but now assume

$$R_1 = R_2 = R_3 = R$$

and

$$d_{\text{proc}} = 0.$$

Suppose that a packet switch does not use store-and-forward. Instead, it immediately begins transmitting each bit as soon as it receives that bit.

Determine the end-to-end delay.

This problem is different from the previous one because the switches do not wait for the entire packet before forwarding it.

This is called **immediate forwarding**.

The important consequence is that transmission over successive links can overlap.

Step 1: Transmission over the First Link

The complete packet takes

$$\frac{L}{R}$$

seconds to be transmitted onto the first link.

The last bit of the packet leaves Host A at time

$$\frac{L}{R}.$$

Step 2: Propagation on the First Link

After being transmitted, the last bit requires

$$\frac{d_1}{s_1}$$

seconds to propagate to the first switch.

Thus, the last bit reaches the first switch at

$$\frac{L}{R} + \frac{d_1}{s_1}.$$

Step 3: Immediate Forwarding

The first switch does not wait for the entire packet.

As soon as a bit is received, it begins transmitting that bit on the second link.

Since the transmission rate is the same on all links,

$$R_1 = R_2 = R_3 = R,$$

there is no additional serialization delay caused by the switch for the last bit.

The last bit therefore propagates through the second link and then the third link.

Consequently, the last bit reaches the destination after the transmission of the packet on the first link plus the propagation delays of all three links:

$$T_{\text{immediate}} = \frac{L}{R} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}.$$

Therefore,

$$T_{\text{immediate}} = \frac{L}{R} + \sum_{i=1}^3 \frac{d_i}{s_i}$$

Why Are There Not Three Transmission Delays?

With store-and-forward, each switch waits for the entire packet. Therefore, the packet must be transmitted completely on every link.

With immediate forwarding, the links operate like a pipeline.

While the first link is still transmitting later bits, the second link can already be transmitting earlier bits, and the third link can also be transmitting bits.

Thus, the transmission delays overlap.

Only one complete packet transmission time,

$$\frac{L}{R},$$

is added to the total propagation delay.

4 Problem 4: Queuing Delay

Problem Statement

A packet switch receives a packet and determines the outbound link. When the packet arrives:

- One packet is already halfway through transmission.
- Four packets are waiting in the queue.

Packets are transmitted in order of arrival.

All packets have length 1500 bytes and the link rate is 2.5 Mbps.

Determine the queuing delay.

Also derive the general expression when:

- Each packet has length L bits.
- Link rate is R bits/sec.
- x bits of the currently transmitted packet have already been transmitted.
- n complete packets are already waiting in the queue.

Queuing delay is the amount of time a newly arriving packet must wait before its transmission begins.

The packet cannot begin transmission until:

1. The remaining part of the currently transmitted packet is sent.
2. The four packets already waiting in the queue are transmitted.

Step 1: Packet Size

The packet size is

1500 bytes.

Therefore,

$$L = 1500 \times 8 = 12,000 \text{ bits.}$$

Thus,

$$\boxed{L = 12,000 \text{ bits}}.$$

Step 2: Transmission Time of One Complete Packet

The link rate is

$$R = 2.5 \text{ Mbps} = 2.5 \times 10^6 \text{ bits/sec.}$$

Therefore,

$$T_{\text{packet}} = \frac{L}{R}.$$

Substituting,

$$T_{\text{packet}} = \frac{12,000}{2.5 \times 10^6}.$$

Thus,

$$T_{\text{packet}} = 0.0048 \text{ sec} = 4.8 \text{ ms.}$$

Hence,

$$\boxed{T_{\text{packet}} = 4.8 \text{ ms}}.$$

Step 3: Remaining Time of the Current Packet

The packet currently being transmitted is halfway complete.

Therefore, half of the packet remains.

The total transmission time is 4.8 ms, so the remaining time is

$$T_{\text{remaining}} = \frac{4.8}{2} = 2.4 \text{ ms.}$$

Therefore,

$$\boxed{T_{\text{remaining}} = 2.4 \text{ ms}}.$$

Step 4: Time for Four Waiting Packets

There are four packets already waiting.

Each requires 4.8 ms to transmit.

Therefore,

$$T_{\text{queue}} = 4(4.8) = 19.2 \text{ ms.}$$

Thus,

$$\boxed{T_{\text{queue}} = 19.2 \text{ ms}}.$$

Step 5: Total Queuing Delay

The arriving packet must wait for:

$$2.4 \text{ ms}$$

for the current packet and

$$19.2 \text{ ms}$$

for the four waiting packets.

Therefore,

$$T_q = 2.4 + 19.2.$$

Hence,

$$T_q = 21.6 \text{ ms}.$$

4.1 General Expression

Suppose:

- Packet length = L bits.
- Link rate = R bits/sec.
- x bits of the current packet have already been transmitted.
- n complete packets are waiting.

The current packet contains L bits.

Since x bits have already been transmitted, the number of remaining bits is

$$L - x.$$

Therefore, the remaining transmission time is

$$\frac{L - x}{R}.$$

Each of the n waiting packets requires

$$\frac{L}{R}$$

seconds.

Thus, the total time required for the n packets is

$$n \frac{L}{R}.$$

Therefore, the queuing delay is

$$T_q = \frac{L - x}{R} + n \frac{L}{R}.$$

Hence,

$$T_q = \frac{L - x + nL}{R}$$

or equivalently,

$$T_q = \frac{(n+1)L - x}{R}.$$

5 Problem 5: Average Queuing Delay

Part (a)

Suppose N packets arrive simultaneously at a link.

Initially, there are no packets being transmitted and no packets waiting.

Each packet has length L and the transmission rate is R .

Determine the average queuing delay.

Since all N packets arrive simultaneously, the first packet can begin transmission immediately.

Therefore, its queuing delay is

$$D_1 = 0.$$

The second packet must wait until the first packet is transmitted:

$$D_2 = \frac{L}{R}.$$

The third packet must wait for two packets:

$$D_3 = 2\frac{L}{R}.$$

Continuing this pattern, the k -th packet has queuing delay

$$D_k = (k - 1)\frac{L}{R}.$$

Therefore, the queuing delays are

$$0, \quad \frac{L}{R}, \quad 2\frac{L}{R}, \quad \dots, \quad (N - 1)\frac{L}{R}.$$

Step 1: Sum of Queuing Delays

The total queuing delay is

$$D_{\text{total}} = \sum_{k=1}^N (k - 1)\frac{L}{R}.$$

Taking L/R outside,

$$D_{\text{total}} = \frac{L}{R} \sum_{k=1}^N (k - 1).$$

The sum is

$$0 + 1 + 2 + \dots + (N - 1).$$

Using

$$0 + 1 + \dots + (N - 1) = \frac{N(N - 1)}{2},$$

we get

$$D_{\text{total}} = \frac{L}{R} \frac{N(N - 1)}{2}.$$

Step 2: Average Queuing Delay

The average is obtained by dividing by N :

$$D_{\text{avg}} = \frac{D_{\text{total}}}{N}.$$

Therefore,

$$D_{\text{avg}} = \frac{1}{N} \frac{L}{R} \frac{N(N - 1)}{2}.$$

Canceling N ,

$$D_{\text{avg}} = \frac{(N - 1)L}{2R}.$$

Hence,

$$\boxed{D_{\text{avg}} = \frac{(N - 1)L}{2R}}.$$

Part (b)

Now suppose N packets arrive every

$$\frac{NL}{R}$$

seconds.

Determine the average queuing delay.

The transmission time for one packet is

$$\frac{L}{R}.$$

Therefore, transmitting N packets requires

$$N \frac{L}{R} = \frac{NL}{R}.$$

This is exactly the interval between two consecutive groups of N arriving packets. Thus, one group of N packets is completely transmitted before the next group arrives. Consequently, each group behaves independently. For every group, the queuing delays are

$$0, \quad \frac{L}{R}, \quad 2\frac{L}{R}, \quad \dots, \quad (N-1)\frac{L}{R}.$$

Therefore, the average queuing delay is the same as in part (a):

$$D_{\text{avg}} = \frac{(N-1)L}{2R}.$$

6 Problem 6: Queuing Delay, Traffic Intensity, and Total Delay

Given Information

Traffic intensity is defined as

$$I = \frac{La}{R},$$

where:

- L = packet length
- a = packet arrival rate
- R = link transmission rate

The queuing delay is given by

$$D_q = \frac{IL}{R(1 - I)},$$

for

$$I < 1.$$

The problem asks:

- Find a formula for the total delay, consisting of queuing delay plus transmission delay.
- Plot total delay as a function of L/R .

Part (a): Formula for Total Delay

The total delay consists of:

$$D_{\text{total}} = D_q + D_{\text{trans}}.$$

The queuing delay is given as

$$D_q = \frac{IL}{R(1 - I)}.$$

The transmission delay of a packet is

$$D_{\text{trans}} = \frac{L}{R}.$$

Therefore,

$$D_{\text{total}} = \frac{IL}{R(1 - I)} + \frac{L}{R}.$$

Step 1: Take $\frac{L}{R}$ Common

We can write

$$D_{\text{total}} = \frac{L}{R} \left(\frac{I}{1-I} + 1 \right).$$

Write 1 using the common denominator:

$$D_{\text{total}} = \frac{L}{R} \left(\frac{I}{1-I} + \frac{1-I}{1-I} \right).$$

Therefore,

$$D_{\text{total}} = \frac{L}{R} \left(\frac{I+1-I}{1-I} \right).$$

Hence,

$$D_{\text{total}} = \frac{L}{R} \frac{1}{1-I}.$$

Therefore,

$$D_{\text{total}} = \frac{L}{R(1-I)}$$

This is the required total delay.

Step 2: Express the Result in Terms of L/R

We are given

$$I = \frac{La}{R}.$$

Let

$$x = \frac{L}{R}.$$

Then

$$I = ax.$$

Substituting into the total-delay formula,

$$D_{\text{total}} = \frac{x}{1-ax}.$$

Therefore,

$$D_{\text{total}} = \frac{x}{1-ax}, \quad x = \frac{L}{R}$$

The condition $I < 1$ becomes

$$ax < 1.$$

Therefore,

$$x < \frac{1}{a}.$$

Thus, the valid range is

$$0 \leq \frac{L}{R} < \frac{1}{a}.$$

Part (b): Plot Total Delay as a Function of L/R

Let

$$x = \frac{L}{R}.$$

Then

$$D_{\text{total}} = \frac{x}{1 - ax}.$$

The function is

$$D(x) = \frac{x}{1 - ax}.$$

The denominator approaches zero as

$$x \rightarrow \frac{1}{a}.$$

Therefore,

$$D(x) \rightarrow \infty \quad \text{as} \quad x \rightarrow \frac{1}{a}^-.$$

Hence, the graph has a vertical asymptote at

$$x = \frac{1}{a}.$$

Since

$$x = \frac{L}{R},$$

the vertical asymptote occurs when

$$\frac{L}{R} = \frac{1}{a}.$$

At this point,

$$I = \frac{La}{R} = 1.$$

The model is valid only for

$$I < 1.$$

Therefore, the graph is plotted only to the left of the asymptote.